Statistics for Social Science

Week 6

Hypothesis Testing II – Association Tests: Chi-square and Correlation

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Outline

- Part 1 – Two categorical variables
  - The crosstab
  - Chi-square test
- Part 2 – Two continuous variables
  - What is a correlation?
  - The scatterplot
  - Pearson correlation
  - Spearman correlation
- Part 3 – Practice

What You Should Learn Today

1 **Two categorical variables → the chi-square test**

Crosstab with row percentages, then χ^2, df and p – and the “expected count” check

2 **Two continuous variables → correlation**

The scatterplot and r: its direction, its strength and p

3 **Pearson or Spearman?**

Roughly normal → Pearson · skewed or ordinal → Spearman (ranks again)

4 **Report it in one sentence**

One sentence for each test – and no causal language: an association is not a cause

Words Matter: New Terms

English	Türkçe	English	Türkçe
association	ilişki	scatterplot	saçılım grafiği
crosstab (contingency table)	çapraz tablo	correlation coefficient (r)	korelasyon katsayısı
observed count	gözlenen frekans	positive / negative relationship	pozitif / negatif ilişki
expected count	beklenen frekans	rank correlation (Spearman's rho)	sıra korelasyonu
chi-square test (χ^2)	ki-kare testi	causation	nedensellik

Use these terms in your quiz answers and in your proposal – in English and in Turkish.

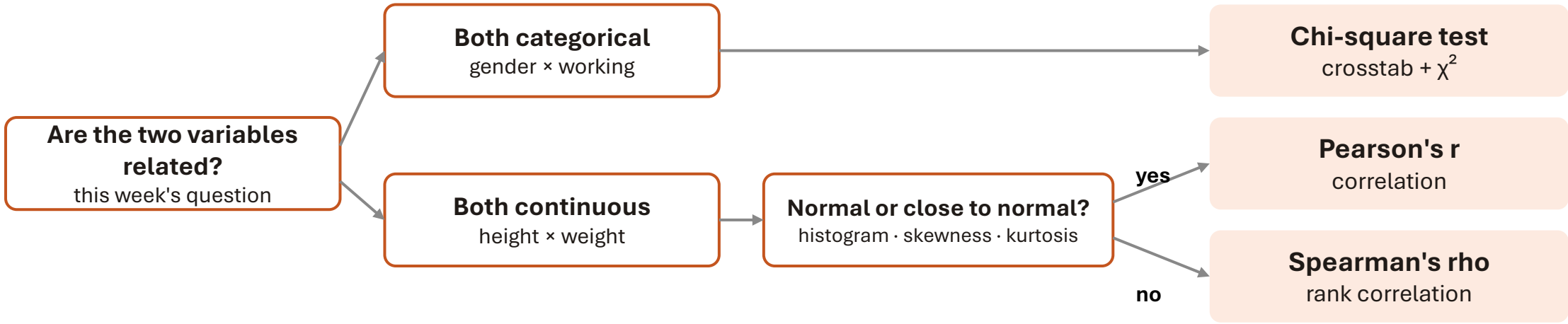
From Groups to Association

LAST WEEK

Does the mean differ between groups? → t-test, Mann–Whitney U, ANOVA, Kruskal–Wallis H

THIS WEEK

Do the two variables go together? Are they related?



The same logic as last week: H_0 says "no association". $p < .05$ → reject H_0.

Two Categorical Variables

The question

Is working for income related to gender?

Variables:

gender – male / female (independent, 2 groups)

empstat1 – works for income: yes / no (dependent, 2 groups).

H_0: gender and working are not related

→ the share of people who work **is the same** for women and men.

H_1: gender and working are related

→ the share of people who work **differs** between women and men.

Both variables are categorical, so there is **no mean**.

We compare **percentages**.

SPSS MENU

Analyze

→ Descriptive Statistics

→ Crosstabs

Row(s): gender

Column(s): empstat1

Cells: tick Observed, Expected, and Row percentages

Statistics: tick Chi-square

Read the Crosstab First

			1,00 İşaretli	2,00 İşaretli değil	Total
gender Cinsiyetiniz nedir?	1,00 Erkek	Count	701	630	1331
		Expected Count	507,5	823,5	1331,0
		% within gender Cinsiyetiniz nedir?	52,7%	47,3%	100,0%
	2,00 Kadın	Count	296	988	1284
		Expected Count	489,5	794,5	1284,0
		% within gender Cinsiyetiniz nedir?	23,1%	76,9%	100,0%
Total		Count	997	1618	2615
		Expected Count	997,0	1618,0	2615,0
		% within gender Cinsiyetiniz nedir?	38,1%	61,9%	100,0%

Columns: 1,00 İşaretli = works for income · 2,00 İşaretli değil = does not work

WHY PERCENTAGES?

The groups have different sizes.

Raw counts mislead: a bigger group has bigger counts.

Percentages put women and men on the same scale.

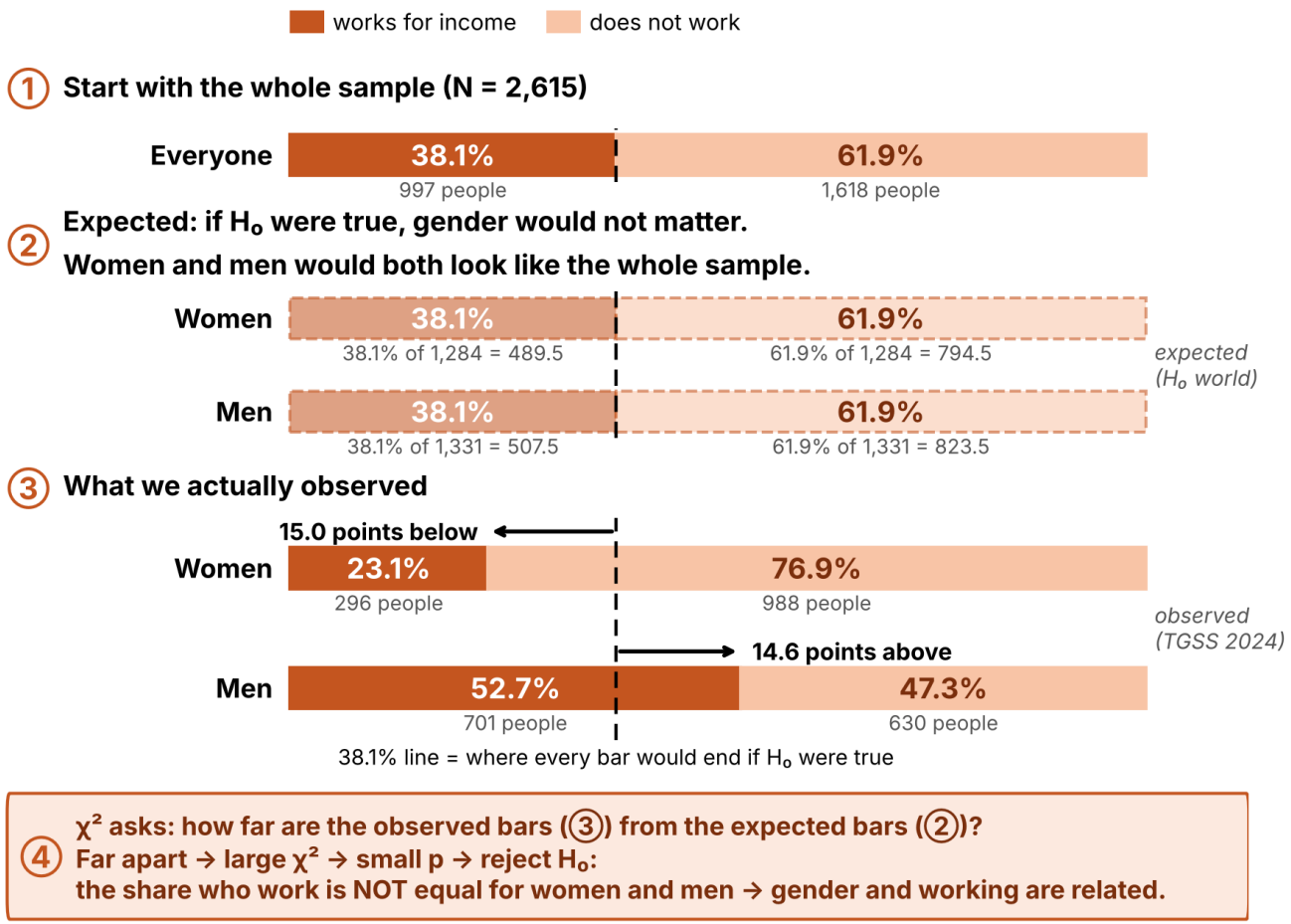
Describe first, then test: the crosstab shows the size and the direction; the test says whether it is more than chance.

The rule: put the independent variable in the rows and ask for row percentages – then you compare like with like.

Men: 52.7% work for income, and the remaining 47.3% do not.
Women: 23.1% work for income, and the remaining 76.9% do not.
Whole sample: 38.1% work for income, and 61.9% do not.
Each row adds up to 100%, so we can compare men and women directly: men work more than women (52.7% vs. 23.1%).

And what is the "Expected Count"?
Expected Count: what we would see if H_0 were true– e.g. 489.5 working women. More on the next slide.

Chi-square: Rule and Logic



Walk through the picture step by step.

(1) The whole sample: 38.1% work.

(2) If H_0 were true, gender would not matter, so women and men would both look like the whole sample – 38.1% work, 61.9% do not. **That is where the expected counts come** from (38.1% of 1,284 women **= 489.5**).

(3) What we observed: women are 15 points below the line, men 14.6 points above.

Chi-square: Output

In the output	What it tells you
Crosstab: row %	Describe first: which group has the higher share?
Pearson Chi-Square: Value, df	The test statistic and its degrees of freedom – both go in the report
Asymp. Sig. (2-sided)	p for H_0: the variables are not related. p < .05 → significant association
Footnote: "… cells have expected count less than 5"	The check: if more than 20% of the cells have an expected count below 5, the test is not reliable → merge categories

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	242,964[a]	1	,000		
Continuity Correction[b]	241,710	1	,000		
Likelihood Ratio	248,397	1	,000		
Fisher's Exact Test				,000	,000
Linear-by-Linear Association	242,871	1	,000		
N of Valid Cases	2615				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 489,54.

b. Computed only for a 2x2 table

WHAT TO LOOK AT

Pearson Chi-Square row

Value (χ^2), df and Asymp. Sig. (p)

p < .05 → the two variables are related

THE LOGIC BEHIND IT

χ^2 adds up the gaps

The bigger the gaps between observed and expected counts, the bigger χ^2 and the smaller p.

df = (rows − 1) × (columns − 1)

REPORT IT IN ONE SENTENCE

Working for income was significantly related to gender: 52.7% of men and 23.1% of women worked, $\chi^2(1, N = 2,615) = 242.96$, p < .001.

What Is a Correlation?

Correlation (r) shows how two variables change together: when x goes up, does y go up, go down, or stay the same?

r is between −1 and +1.

r > 0 (positive): they change in the same direction – x ↑, y ↑.

r < 0 (negative): they change in opposite directions – e.g. warmer weather ↑, gas bill ↓.

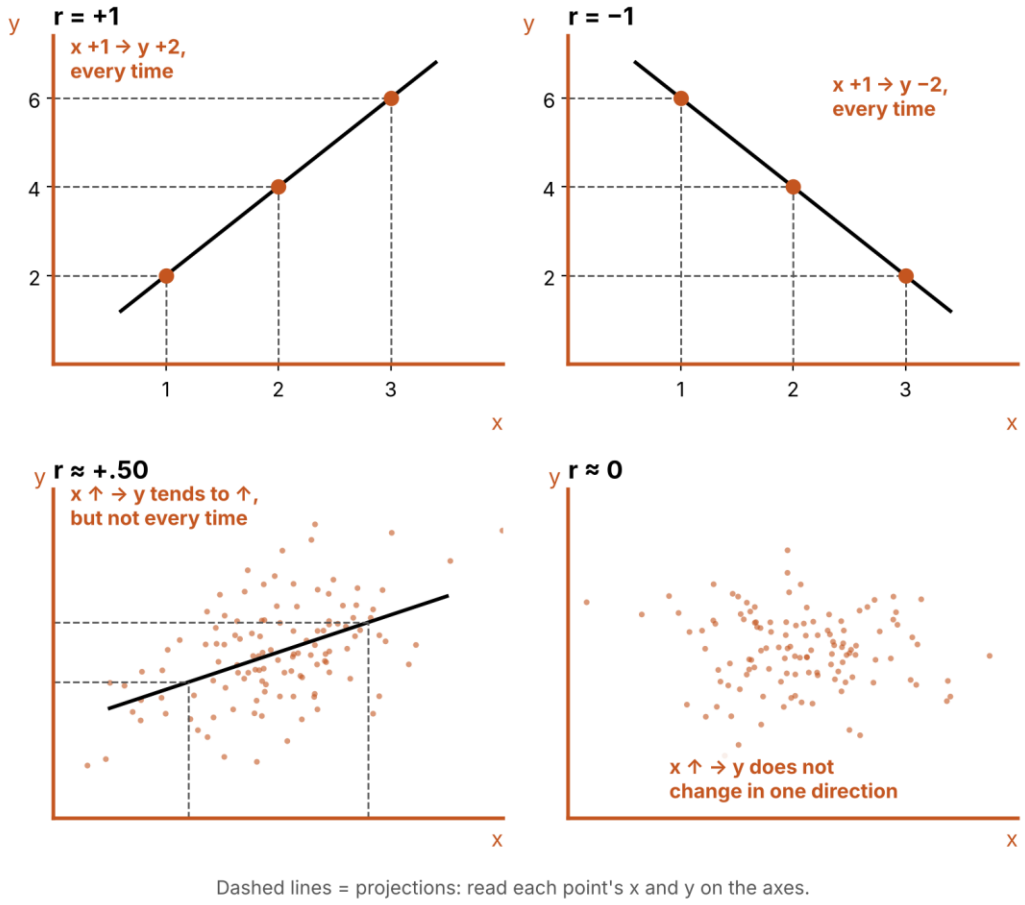
Strength: closer to ±1 → stronger; closer to 0 → weaker. r = 0 → no linear relationship.

r = ±1 does not mean “y changes by 1”. It means y changes by a fixed amount every time x changes (here 2).

Top left (r = +1): a positive linear relationship. As x goes up, y goes up too. Every time x increases by 1, y increases by 2.

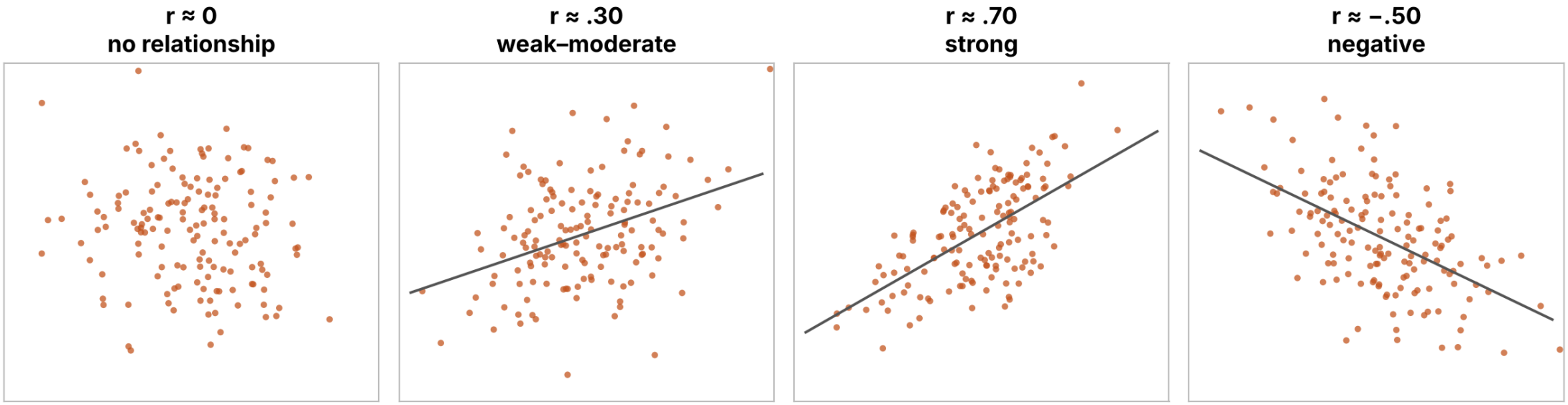
Top right (r = −1): a negative linear relationship. As x goes up, y goes down – or the other way round: as x goes down, y goes up. Every time x increases by 1, y decreases by 2.

Linear means that a change in one variable goes together with a change of a fixed amount in the other, so the points follow a straight line.



Dashed lines = projections: read each point's x and y on the axes.

Two Continuous Variables: The Scatterplot



Simulated data, 150 people each

Every point is one person. Look at the direction (up or down), the strength (how close to a line) and the form (is it a straight line?).

Pearson's r

r = Pearson's correlation coefficient: one number, from −1 to +1, that shows the direction (its sign) and the strength (its size) of a linear relationship.

What it measures: the linear relationship between two continuous variables whose distributions are normal or close to normal.

The sign = direction:

+ both go up together;

– one goes up, the other goes down.

The size = strength (rule of thumb; Cohen, 1988, in Field): ±.10 small · ±.30 medium · ±.50 large.

NO DEPENDENT, NO INDEPENDENT

Both variables are treated the same.

The correlation of height with weight = the correlation of weight with height.

CAREFUL

Correlation is not causation.

We don't know which one causes which – or whether either one does.

A third variable can create the association (e.g. age).

Write "related to" / "associated with", not "causes" / "affects".

H_0: The correlation between height and weight in the population is equal to 0 → there is no linear relationship.

H_1: The correlation in the population is not equal to 0 → there is a linear relationship.

In SPSS: Scatterplot and Pearson's r

1 Draw the scatter plot

Graphs → Legacy Dialogs → Scatter/Dot → Simple Scatter
→ Define. Y Axis: weight · X Axis: height → OK.

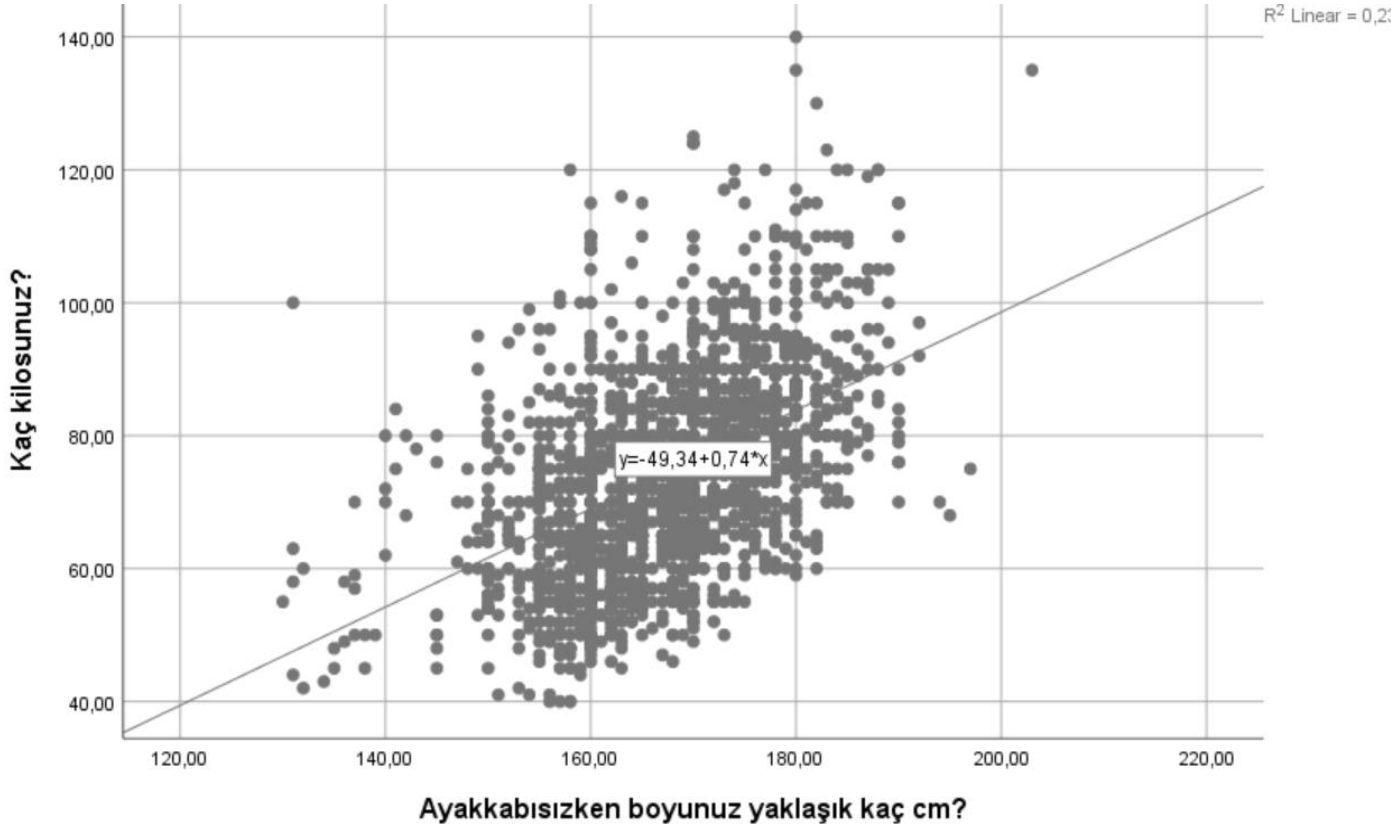
2 Add the trend line

Double-click the chart → Elements → Fit Line at Total.

3 Test it

Analyze → Correlate → Bivariate
→ move both variables
→ tick Pearson → OK.

Which variable on which axis? For a correlation it does not matter.



Reading the Correlation Output

Correlations

		height Ayakkabısızken boyunuz yaklaşık kaç cm?	weight Kaç kilosunuz?
height Ayakkabısızken boyunuz yaklaşık kaç cm?	Pearson Correlation	1	,482**
	Sig. (2-tailed)		,000
	N	1701	1694
weight Kaç kilosunuz?	Pearson Correlation	,482**	1
	Sig. (2-tailed)	,000	
	N	1694	1709

**. Correlation is significant at the 0.01 level (2-tailed).

In the output	What it tells you
Pearson Correlation	r: direction and strength
Sig. (2-tailed)	p < .05 → significant correlation
N	Number of people with both values
The table is a mirror	The diagonal is always 1 (a variable with itself); the two halves repeat each other – read one

REPORT IT IN ONE SENTENCE

Height and weight were positively correlated, r(1692) = .48, p < .001 – a medium-to-large correlation.

Spearman's rho: Ranks Again

When?

Spearman is the nonparametric version of Pearson. Use it when one or both variables are far from normal (clearly skewed), or ordinal (e.g. a 1–5 scale).

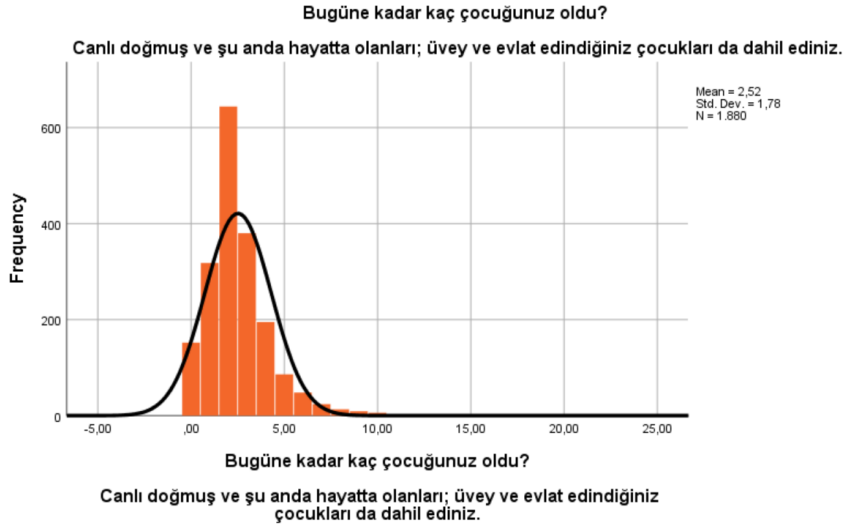
How?

Rank each variable from the smallest to the largest, then correlate the ranks – the same idea as Mann–Whitney last week. Extreme values lose their pull.

Our example

childs × childideal: the number of children people have, and the number they think is ideal. Both are counts and skewed to the right (skewness 2.08 and 2.24).





Spearman's rho in SPSS

Correlations

			childdideal Sizce bir ailenin sahip olması gereken ideal çocuk sayısı nedir?	childs Bugüne kadar kaç çocuğunuz oldu? Canlı doğmuş ve şu anda hayatta olanları; üvey ve evlat edindiğiniz çocukları da dahil ediniz.
Spearman's rho	childdideal Sizce bir ailenin sahip olması gereken ideal çocuk sayısı nedir?	Correlation Coefficient	1,000	,392**
		Sig. (2-tailed)	.	,000
		N	1721	1233
	childs Bugüne kadar kaç çocuğunuz oldu? Canlı doğmuş ve şu anda hayatta olanları; üvey ve evlat edindiğiniz çocukları da dahil ediniz.	Correlation Coefficient	,392**	1,000
		Sig. (2-tailed)	,000	.
		N	1233	1880

**. Correlation is significant at the 0.01 level (2-tailed).

SPSS MENU

Analyze → Correlate → Bivariate

Variables: childs, childdideal · tick Spearman

H_0: The association between the ranks of the two variables is equal to 0 in the population.

H_1: The association between the ranks of the two variables is not equal to 0 in the population.

REPORT IT

The number of children people had was positively correlated with the number they considered ideal, rs(1231) = .39, p < .001.

Read the row "Correlation Coefficient" under Spearman's rho.

What You Can Do Now

1 **Read a crosstab**

Independent variable in the rows · row percentages · compare the rows

2 **Run and read a chi-square test**

χ^2, df, p · check the expected counts footnote

3 **Read a scatterplot and a correlation**

Direction, strength, p · Pearson for roughly normal, Spearman for skewed or ordinal

4 **Report without causal language**

“related to”, “associated with” – not “causes”

Part 3: Practice

- Open TGSS 2024 and your syntax file from class
- Work through the Week 6 worksheet – one task per test
- Save your syntax as W6_name.sps
- Ask as you go – I will come around